TITLE: **How fast does the central limit theorem work? Exact distances, simulation noise and heavy tails**

SUBJECT: Aim, method and results of the analyses in version 2 of the `CLT-speed-race-exercise` repository

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1. Aim and scope

Let $X_1, X_2, \ldots$ be independent draws from a distribution with mean μ and variance σ^2, and let $Z_n = \sqrt{n}\,(\bar{X}_n - \mu)/\sigma$ be the standardized sample mean, with distribution function F_n. If $0 < \sigma^2 < \infty$, the central limit theorem (CLT) says that $F_n(x) \to \Phi(x)$ for every x. How fast this happens is measured here by the Kolmogorov distance

$$D_n = \sup_x |F_n(x) - \Phi(x)|,$$

the largest error in any probability $P(Z_n \le x)$ computed from the normal approximation. Version 1 of this project animated simulations of Z_n with $R = 2000$ replications for ten distributions and grouped them by eye: fast (Normal, Uniform), slower (Exponential, Laplace), very slow (Student-t with few degrees of freedom), and failing (Pareto with tail index ≤ 2, Cauchy). This note asks three questions:

1. **How large is D_n exactly, and what determines how fast it falls?** Six of the distributions allow an exact computation.
2. **What can a simulation with $R = 2000$ replications measure?** Even an exactly normal Z_n gives a simulated distance above zero.
3. **What happens for the remaining distributions:** Lognormal, F and Pareto, which have no exact method here, and Cauchy and Pareto with infinite variance, where version 1 stopped?

All distributions use the default parameters of version 1: Uniform(0, 1), Exponential(1), Laplace(0, 1), Lognormal with $\sigma = 0.6$, Chi-square(5), F(5, 10), Student-t(5), Pareto with tail index 2.5. Student-t(3) is added as the smallest integer number of degrees of freedom with finite variance.

2. Method

- **Exact distances.** For Exponential and Chi-square the sum of n draws is gamma distributed, so F_n is a regularized incomplete gamma function. For Uniform with $n \le 40$ the sum has the Irwin–Hall distribution, evaluated in exact rational arithmetic. For the other

cases (Laplace, Uniform with $n > 40$, Student-t), $F_n - \Phi$ is obtained by inverting the difference of the two characteristic functions with the Gil-Pelaez (1951) formula and Gauss–Legendre quadrature. Against the exact methods, and against closed forms for Laplace ($n = 2$) and Student-t ($n = 1$), this inversion agrees to within 3×10^{-15}. The supremum is located on a grid of step 0.02 on $[-8, 8]$ and refined numerically.

- **Leading terms.** For a continuous distribution with finite fourth moment, the Edgeworth expansion (Hall, 1992) gives $F_n(x) - \Phi(x) = -\varphi(x)\left[\frac{\gamma}{6\sqrt{n}} He_2(x) + \frac{\kappa_4}{24n} He_3(x) + \frac{\gamma^2}{72n} He_5(x)\right] + o(1/n)$, with skewness γ, excess kurtosis κ_4 and Hermite polynomials He_k. For $\gamma \neq 0$ the leading term of D_n is $|\gamma|\varphi(0)/(6\sqrt{n}) = 0.0665\,|\gamma|/\sqrt{n}$; for $\gamma = 0$ it is $|\kappa_4| \cdot 0.5506/(24n)$, where $0.5506 = \max_x |He_3(x)\varphi(x)|$. Student-t(3) has no finite third absolute moment. Its standardized characteristic function is $(1 + |s|)e^{-|s|} = 1 - s^2/2 + |s|^3/3 + O(s^4)$, and the term $|s|^3/3$ gives the leading term $0.1100/\sqrt{n}$ (derived for this note and computed numerically).
- **Simulation.** The simulation of version 1 is repeated draw for draw (same seed, same calls of the random number generator); it reproduces four distances printed in the frames that version 1 saved. The noise of a simulation is measured by the Kolmogorov statistic of R draws from the true distribution, whose exact distribution is available in SciPy (`scipy.stats.kstwo`).
- **Checks.** Seventeen unit tests compare the code with independent computations: moments with SciPy, characteristic functions with numerical Fourier integrals of the densities, the inversion with exact distribution functions, the supremum with a brute-force search.

3. Analysis 1: The exact speed of convergence (Figure 1, Table 1)

Why. A ranking of distributions by eye cannot say how much faster one converges than another, or why. The exact D_n answers the first question, and the comparison with the leading terms answers the second.

Method. D_n for 41 values of n from 1 to 10^4, evenly spaced on a log scale, for six distributions. For each, the smallest n with $D_n < 0.01$, found by computing D_n for every n up to that point and checking the following 200 values.

How to read the figure. Left: both axes are logarithmic, so a distance proportional to $n^{-1/2}$ is a line of slope $-1/2$, and one proportional to n^{-1} a line of slope -1. Dashed lines are the leading terms. Right: the ratio of D_n to its leading term; a curve that approaches 1 means that the leading term explains the speed of convergence.

Finding. The distributions fall into two groups. For the symmetric distributions with a finite fourth moment (Uniform, Laplace, Student-

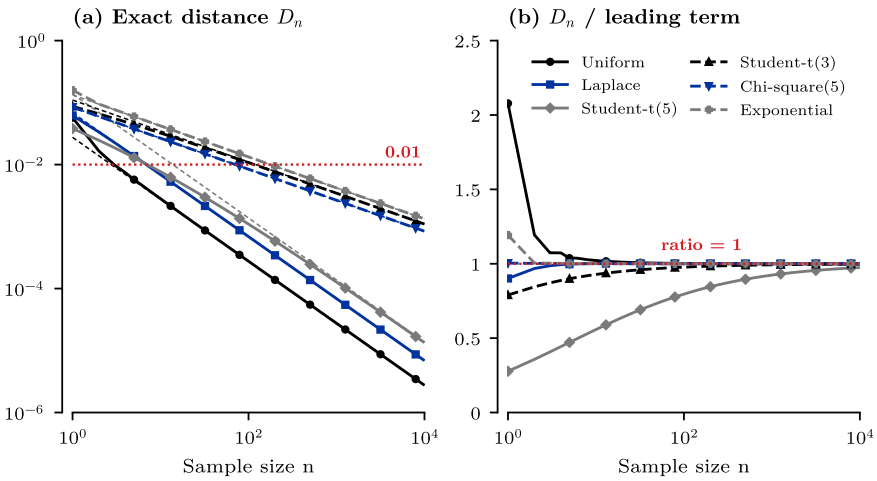


Figure 1. Exact Kolmogorov distance between the standardized mean and the normal distribution. Left: D_n (solid) and its leading term (dashed); dotted: 0.01. Right: D_n divided by its leading term.

t(5)) D_n falls like $1/n$; for the skewed ones (Chi-square(5), Exponential) only like $1/\sqrt{n}$. Student-t(3) is symmetric but also falls like $1/\sqrt{n}$. At $n = 10^4$ the ratio to the leading term is 1.000 (Uniform), 1.000 (Laplace), 1.000 (Chi-square), 1.000 (Exponential) and 0.998 (Student-t(3)); for Student-t(5) it approaches 1 more slowly (0.974). The leading term predicts the smallest n with $D_n < 0.01$ exactly for Uniform, Laplace, Chi-square and Exponential ($n = 3$, 7, 71, 177). For Lognormal(0.6) and F(5, 10) it predicts $n \approx 226$ and 662. At $n = 30$, Exponential is still at $D_{30} = 0.0243$, 10.6 times Laplace (0.00230).

Interpretation. The speed of convergence is set mainly by the skewness: it decides whether the error falls like $1/\sqrt{n}$ or like $1/n$. Heavier tails as such, as long as the fourth moment is finite, do not change the rate: Laplace has heavier tails than the uniform distribution and converges at the same rate, and version 1 placed it wrongly next to Exponential. What slows Student-t(3) is the infinite third absolute moment, which adds a $1/\sqrt{n}$ term without any skewness; Student-t(5), the default of version 1, is already at $D_n < 0.01$ for $n = 8$. A single rule such as "$n \geq 30$" cannot describe this: among the distributions of version 1 the n needed for $D_n < 0.01$ ranges from 3 (Uniform) to about 662 (F(5, 10), leading-term estimate), and Pareto(2.5) has not reached it at $n = 10^4$ (analysis 3).

4. Analysis 2: What a simulation with 2000 replications measures (Figure 2)

Why. Version 1 measured the distance from $R = 2000$ simulated values of Z_n, as $\max|\hat{F}(x) - \Phi(x)|$ over 401 grid points on $[-12, 12]$. With finite

Table 1. Skewness γ, excess kurtosis κ_4, leading term of D_n, exact D_n, and the smallest n with $D_n < 0.01$, exact and as predicted by the leading term. –: does not exist, or no exact method here.

Distribution	γ	κ_4	Leading term	D_{30}	D_{1000}	n for $D_n < 0.01$ exact	leading term
Normal	0	0	0	0	0	1	–
Uniform	0	−1.20	$0.0275\,n^{-1}$	9.24×10^{-4}	2.75×10^{-5}	3	3
Laplace	0	3.00	$0.0688\,n^{-1}$	0.00230	6.88×10^{-5}	7	7
Student-t(5)	0	6.00	$0.1376\,n^{-1}$	0.00314	1.27×10^{-4}	8	14
Student-t(3)	–	–	$0.1100\,n^{-1/2}$	0.0192	0.00345	116	121
Chi-square(5)	1.26	2.40	$0.0841\,n^{-1/2}$	0.0154	0.00266	71	71
Exponential	2.00	6.00	$0.1330\,n^{-1/2}$	0.0243	0.00421	177	177
Lognormal(0.6)	2.26	10.3	$0.1503\,n^{-1/2}$	–	–	–	226
F(5, 10)	3.87	50.9	$0.2571\,n^{-1/2}$	–	–	–	662
Pareto(2.5)	–	–	–	–	–	–	–
Cauchy	no mean or variance						

R, $\hat{F}$ differs from F_n even when $F_n = \Phi$. The simulated distance can show convergence only while D_n is above this noise.

Method. The simulation of version 1 for Normal, Uniform and Exponential (seed 42, 30 sample sizes from 10 to 10^4), compared with the exact D_n. The mean noise for $R = 2000$ is 0.0193 for the exact empirical distance (95% quantile 0.0303; the asymptotic value $\sqrt{\pi/2}\,\ln 2/\sqrt{R}$ is 0.0194) and 0.0171 for the grid distance of version 1 (400 simulated samples). Right panel: the number of replications at which the mean noise, $0.8687/\sqrt{R}$, equals D_n.

How to read the figure. Left: markers are the distances of version 1, lines the exact D_n; the shaded region lies below the mean noise. Right: the dotted lines mark the default and the largest R that version 1 offered.

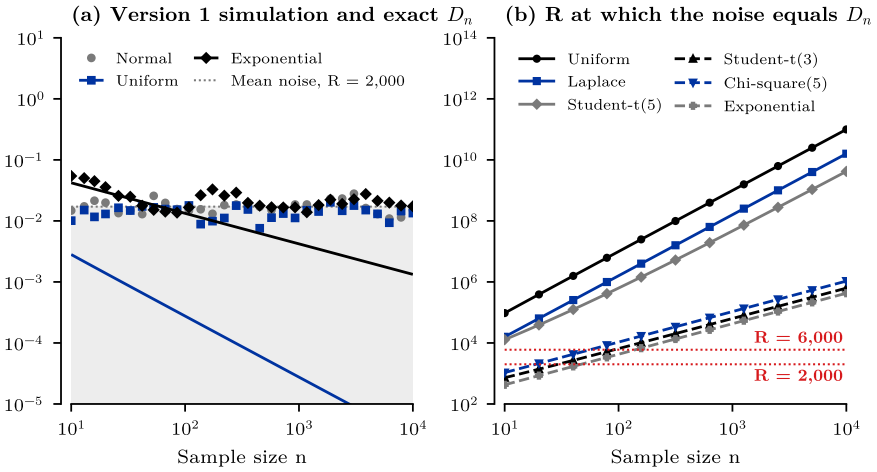


Figure 2. Left: simulated distances of version 1 (markers; $R = 2000$, seed 42) and exact D_n (lines); dotted: mean noise of the grid distance of version 1. Right: replications needed for the mean noise to equal D_n.

Finding. For the Normal distribution Z_n is exactly normal, yet version 1 reports 0.0151 at $n = 10^4$; for Uniform it reports 0.0134, while the exact distance is 2.75×10^{-6}. The exact distance falls below the mean noise of the exact empirical distance (0.0193) from $n = 48$ for Exponential and from $n = 2$ for Uniform. Of the 30 frames of each run, the exact distance was below the noise of version 1 in 30 for Uniform, Laplace and Student-t(5), in 26 for Chi-square(5) and in 22 for Exponential. To make the mean noise as small as D_n, Uniform needs $R \approx 8.8 \times 10^5$ replications at $n = 30$ and 1.0×10^9 at $n = 1000$; Exponential needs 1,279 and 42,676. Version 1 also drew the difference $\hat{F} - \Phi$ with fixed limits of about ± 0.055; for Pareto(2.5) the distance exceeded them in 19 of 30 frames (up to 0.151).

Interpretation. Most frames of the race of version 1 show simulation noise, not convergence: from a few dozen draws on, every distribution of the race except Pareto(2.5) looks equally close to normal. A simulation can only resolve distances larger than about $0.87/\sqrt{R}$, so differences between symmetric distributions at moderate n would need millions to billions of replications. Exact computation is the only practical way to compare them.

5. Analysis 3: Heavy tails (Figure 3)

Why. Two questions are left from version 1: how fast Lognormal, F and Pareto(2.5) converge, which have no exact method here, and what the sample mean does when the variance is infinite, where version 1 only printed a message.

Method. (a) Interquartile range (IQR) of $\bar{X}_n$, which exists even without a finite variance: Pareto with tail indices 2.5 and 1.5 and Cauchy, n from 1 to 10^4, $R = 20{,}000$ paths (100,000 for Pareto(2.5)); for comparison the exact IQR $2 \cdot 0.6745/\sqrt{n}$ of Normal(0, 1). (b) The simulated distance of Z_n for Lognormal(0.6), F(5, 10) and Pareto(2.5) with $R = 100{,}000$ paths (mean noise 0.0027). Slopes are fitted on the log–log scale for $n \geq 10$ where the simulated distance exceeds three times the noise.

How to read the figure. Both panels are logarithmic. Left: a slope of $-1/2$ is the $1/\sqrt{n}$ shrinking of the CLT; a flat line means that averaging does not reduce the spread. Right: dashed lines are the leading terms where the skewness exists.

Finding. The IQR of the Cauchy mean is 1.98 at $n = 1$ and 2.01 at $n = 10^4$ (slope -0.006): the mean of n Cauchy variables has the same Cauchy distribution as one of them. For Pareto(1.5), which has a mean but no variance, the slope for $n \geq 100$ is -0.30, close to $1/\alpha - 1 = -1/3$ from the generalized CLT (Gnedenko and Kolmogorov, 1954); for Pareto(2.5) it is -0.46, still approaching $-1/2$. For Lognormal(0.6) and F(5, 10) the simulated distance falls with slopes -0.42 and -0.40; at

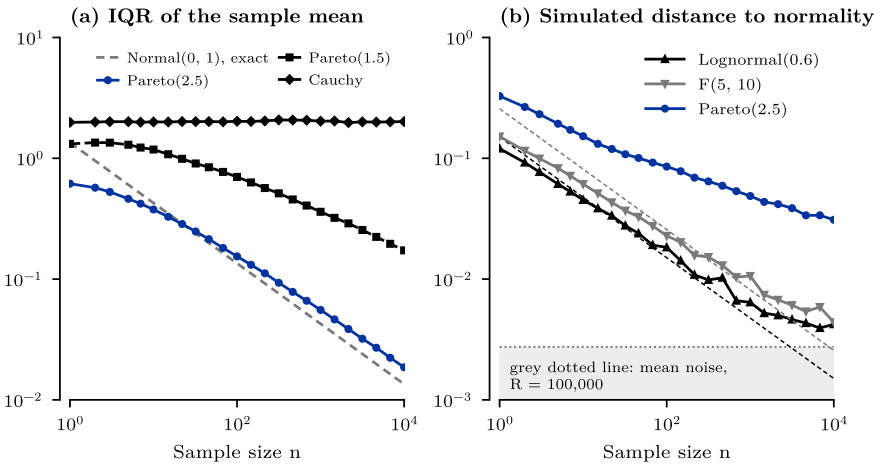


Figure 3. Left: IQR of the sample mean. Right: simulated distance of Z_n to the normal distribution (R = 100,000), with leading terms (dashed); shaded: below the mean noise.

$n = 10$ it is 0.045 against a leading term of 0.048 (Lognormal) and 0.061 against 0.081 (F). Pareto(2.5), whose third moment is infinite, converges much more slowly: 0.152, 0.085, 0.049 and 0.031 at $n = 10$, 10^2, 10^3, 10^4 (slope -0.23). At $n = 10^4$ it is further from normal than Exponential at $n = 30$.

Interpretation. With infinite variance the spread of the sample mean does not shrink like $1/\sqrt{n}$. For Cauchy averaging does not reduce it at all. For Pareto(1.5) the mean still converges to μ, but at the rate $n^{1/\alpha-1}$ and with a non-normal (stable) limit, so a histogram standardized with $\sqrt{n}$ cannot settle. Finite variance alone guarantees convergence but not its speed. The Lyapunov–Katz bound (Katz, 1963) states $D_n \leq C\,\nu_{2+\delta}\,n^{-\delta/2}$, $0 < \delta \leq 1$, where $\nu_{2+\delta} = E|Y|^{2+\delta}$ for the standardized variable Y. For Pareto(2.5) only moments of order below 2.5 exist, so the bound applies only with $\delta < 0.5$ and guarantees a decrease only at a rate of up to about $n^{-1/4}$. The measured slope, -0.23, is of this order.

6. Conclusions and limitations

- The speed of convergence is governed mainly by the skewness: D_n falls like $1/n$ for symmetric distributions with a finite fourth moment and like $1/\sqrt{n}$ for skewed ones, with the constant given by the leading term of the Edgeworth expansion. Infinite third moments (Student-t(3), Pareto(2.5)) slow convergence further.
- A simulation with R replications cannot resolve distances below about $0.87/\sqrt{R}$. Most frames of the race of version 1 show this noise.
- With infinite variance the CLT in its usual form does not apply: the Cauchy mean does not concentrate, and the Pareto(1.5) mean concentrates like $n^{-1/3}$.

- The Kolmogorov distance measures the largest error in a probability, not the relative error in the tails, which can be much larger. Lognormal, F and Pareto are studied by simulation only. Only the default parameters of version 1 are used.

References

Gil-Pelaez, J. (1951). Note on the inversion theorem. *Biometrika*, 38(3–4), 481–482.

Gnedenko, B. V., & Kolmogorov, A. N. (1954). *Limit Distributions for Sums of Independent Random Variables*. Addison-Wesley.

Hall, P. (1992). *The Bootstrap and Edgeworth Expansion*. Springer.

Katz, M. L. (1963). Note on the Berry–Esseen theorem. *Annals of Mathematical Statistics*, 34(3), 1107–1108.

SciPy documentation: `scipy.special.gammainc`, `scipy.stats.kstwo`.

All numbers are generated from `figures/*.csv` by `make_paper.py`.